

An Empirical Study of Gap-Aware LSTM Pipelines for Weather Forecasting: A Case in Ho Chi Minh City

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Abstract

Multivariate time series weather forecasting using operational API archives frequently encounters missing timesteps and irregular sampling, which corrupts temporal alignment and cyclical feature encoding. In this work, we present an empirical evaluation of recurrent neural network preprocessing pipelines for single-step temperature forecasting using historical meteorological data collected from Ho Chi Minh City via the OpenWeatherMap API over a 647-day period. We compare a baseline pipeline (LSTMv0.0.9-beta), which assumes strict three-hour step continuity, against a gap-aware pipeline (LSTMv0.0.9-alpha), which integrates fixed-grid reindexing, a missingness mask, a delta-time feature, physically-aware solar radiance interpolation, and large-gap window filtering. Both pipelines utilize an identical backbone consisting of a two-layer Long Short-Term Memory (LSTM) network trained with Huber loss. Evaluated on a complete 346-day dataset (zero missing rows) and a sparse 645-day dataset (86 scattered missing rows), LSTMv0.0.9-alpha achieves a lower test loss of 0.0012 and a MAPE of 2.33% compared to 0.0014 and 2.59% for the baseline on sparse data, while maintaining equivalent performance on clean data (0.0018 test loss). Our empirical results demonstrate that explicitly handling temporal gaps and enforcing physical domain constraints greatly improves model robustness and generalization in long-term, imperfect forecast archives.

1 Introduction

Accurate time series weather forecasting plays a fundamental role in urban planning, energy grid management, alongside disaster risk reduction.

Over the past decade, deep learning models—especially Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) architectures—have become standard baselines for modeling long-term temporal dependencies across multivariate meteorological variables such as temperature and humidity, atmospheric pressure, and wind speed.

However, deploying weather prediction models in real-world environments poses major data quality challenges. Meteorological data archived from public APIs (e.g., OpenWeatherMap) frequently suffers from missing timesteps and irregular sampling caused by API rate limiting, network interruptions, or sensor outages.

Standard time series forecasting pipelines often assume perfect temporal continuity and generate synthetic index-based cyclical encodings (e.g., daily and seasonal sine/cosine features). When applied to sparse archives containing temporal gaps, these naive baseline models misalign cyclical time features relative to true calendar time, degrading temporal fidelity and prediction accuracy.

While complex theoretical frameworks such as GRU-D (which incorporates trainable hidden-state decay rates) and CRIB (which utilizes an Information Bottleneck to bypass imputation) have been proposed to handle missing data, lightweight data-engineering pipelines built around standard LSTM backbones remain highly desirable for practical deployment due to their computational efficiency and ease of implementation.

This paper presents an empirical case study centered on a 647-day meteorological forecast archive (September 17, 2024 to June 25, 2026) collected for Ho Chi Minh City. The data is sampled at 3-hour intervals, yielding an expected frequency of 8 timesteps per day.

We design and compare two distinct preprocessing pipelines:

- **LSTMv0.0.9-beta (Baseline):** Assumes uniform 3-hour spacing and uses row-position-based cyclical time encoding without missingness handling or data interpolation.
- **LSTMv0.0.9-alpha (Gap-Aware Pipeline):** Re-indexes timestamps onto a strict 3-hour grid, tracking missing observations via a binary missingness mask and a delta-time feature. It applies linear interpolation for general meteorological variables while enforcing physical rules for solar radiance (forcing nighttime solar values to zero) and filters out sequence windows containing gaps exceeding 12 hours.

Primary contributions The key contributions of this empirical report are summarized as follows:

- **Dataset curation & open release:** We compile and publish a 647-day multi-feature weather archive (5,152 total timesteps) for Ho Chi Minh City, made available on Kaggle to support reproducible research on time series modeling under real-world data gaps.
- **Domain-informed pipeline design:** We formulate a gap-aware preprocessing architecture (LSTMv0.0.9-alpha) that combines domain-specific physical constraints (nighttime solar radiance forcing) with temporal gap encodings without altering the underlying standard LSTM network structure.
- **Controlled empirical evaluation:** Through comparative experiments on complete (346 days, 0 missing rows) and sparse (645 days, 86 missing rows) archives, we show that while both pipelines perform identically on clean datasets, the gap-aware pipeline reduces MAPE (2.33% vs. 2.59%) and test loss (0.0012 vs. 0.0014) on incomplete real-world archives, proving that proper gap preprocessing preserves temporal fidelity and improves model generalization.

2 Related Work

Recurrent neural networks for time series. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks were established to model sequential dynamics and long-term temporal dependencies through gating mechanisms that mitigate vanishing and exploding gradients [Hochreiter and Schmidhuber, 1997]. Standard LSTM architectures remain fundamental baselines for multivariate time series forecasting across diverse operational domains, including meteorology, traffic monitoring, and also medical health systems.

Multivariate time series forecasting with missing values. Historically, multivariate time series forecasting with missing values has been approached via two-stage “imputation-then-prediction” paradigms, where missing observations are first filled using traditional statistical techniques (e.g., mean, forward fill, spline interpolation) or learning-based imputation models (e.g., BRITS, SAITS, TimesNet) before executing downstream forecasting. However, recent empirical analysis by Yang et al. [2025] demonstrates that ungrounded imputation without direct ground-truth supervision can distort the underlying data distribution and corrupt inter-variate correlations, causing error propagation that actively harms prediction performance.

Direct modeling of missing patterns. To circumvent flawed imputation, specialized architectures like GRU-D incorporate informative missingness directly into recurrent cell dynamics using masking vectors and time-interval decay functions [Che et al., 2018]. Similarly, direct-prediction frameworks such as CRIB utilize Information Bottleneck principles and attention mechanisms to filter missingness noise without explicitly reconstructing unobserved values. While complex specialized models require modifying internal network equations, lightweight data-engineering approaches—such as passing delta-time features and missingness masks into standard LSTM backbones—provide a practical, deployment-friendly alternative for operational forecast archives.

3 Dataset and Problem Formulation

3.1 Dataset curation & meteorological features

The dataset evaluated in this empirical study consists of archived 5-day / 3-hour forecast data retrieved from the OpenWeatherMap API for **Ho Chi Minh City**.

The complete archive spans 647 days from September 17, 2024, to June 25, 2026, with a sampling interval of 3 hours (yielding an expected frequency of 8 timesteps per day).

The input features are structured into three functional groups:

- **Base meteorological features (6):** Temperature, humidity, wind speed, atmospheric pressure, cloudiness, and solar radiance.
- **Auxiliary temporal features (4):** Cyclical sine and cosine encodings for the hour of the day ($\sin_{\text{hour}}, \cos_{\text{hour}}$) and day of the year ($\sin_{\text{day}}, \cos_{\text{day}}$) to preserve daily and seasonal cycles.
- **Robustness features (Alpha pipeline only, 2):** A binary missingness mask indicating whether a timestep was observed (0) or imputed (1), and a delta-time feature (Δt) representing elapsed time since the last valid observation.

To evaluate performance under clean versus sparse real-world conditions, two specific experimental subsets are compiled:

- **Dataset (1) — Complete data:** Spans 346 days (January 12, 2025, to December 23, 2025), comprising 2,760 consecutive timesteps with zero missing rows. This dataset isolates model performance under ideal, uninterrupted sampling conditions.
- **Dataset (2) — Sparse data:** Spans 645 days (September 18, 2024, to June 24, 2026), comprising 5,152 total timesteps with 86 missing rows scattered throughout the archive. This dataset reflects real-world operational API collection failures and irregular gap distribution.

3.2 Problem formulation

Given a multivariate historical sequence $X \in \mathbb{R}^{T \times D}$ over a lookback window of $T = 72$ timesteps (corresponding to 9 days of 3-hour observations) across D features, the target is to predict the single-step temperature forecast at timestep $T + 1$. The dataset is split into an 80/20 train/test ratio.

4 Methodology

This section outlines the architectural backbone and contrasts the two preprocessing pipelines: the baseline naive pipeline (LSTMv0.0.9-beta) and the gap-aware pipeline (LSTMv0.0.9-alpha).

4.1 Architectural backbone

Both pipeline variants share an identical recurrent neural network backbone to isolate the impact of data preprocessing.

The core model consists of a two-layer Long Short-Term Memory (LSTM) network with a hidden dimension of 128 units per layer and a dropout rate of 0.2 applied between layers to prevent overfitting:

$$h_t, c_t = \text{LSTM}(x_t, h_{t-1}, c_{t-1}) \quad (1)$$

The input sequence length is set to $T = 72$ timesteps (representing 9 consecutive days of 3-hour observations). The output layer projects the final recurrent hidden state h_T through a dense linear layer to produce a single-step point forecast for temperature at timestep $T + 1$.

Training is conducted using mini-batch optimization with the Huber loss function, which combines squared-error and absolute-error behaviors to maintain robustness against meteorological extreme values and outliers:

$$L_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta \\ \delta (|y - \hat{y}| - \frac{1}{2}\delta) & \text{otherwise} \end{cases} \quad (2)$$

The optimization setup includes the Adam optimizer, an automatic learning rate scheduler, gradient clipping to prevent exploding recurrent gradients, early stopping based on validation loss, and finally model checkpointing to preserve the best parameters.

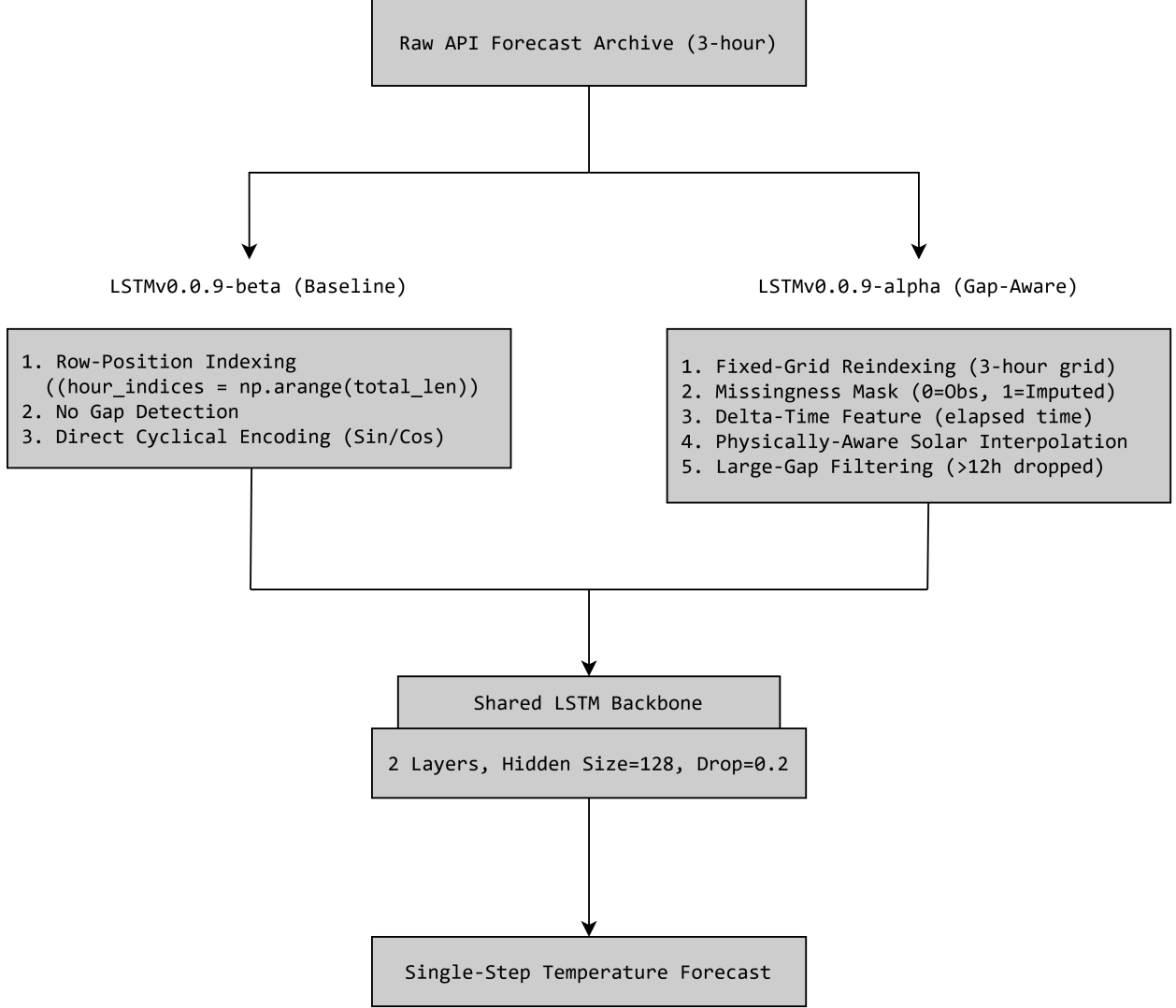


Figure 1: Architecture of the pipeline

4.2 Baseline preprocessing pipeline: LSTMv0.0.9-beta

The baseline pipeline, designated as LSTMv0.0.9-beta, represents a standard time series forecasting setup that assumes perfect, uninterrupted temporal sampling.

- **Uniform spacing assumption:** Beta assumes that every consecutive record in the dataset is separated by exactly 3 hours without checking timestamp continuity.
- **Synthetic index encoding:** To generate cyclical time features, Beta constructs synthetic time indices derived directly from row position indices instead of true calendar timestamps:

$$\text{hour_indices} = 0, 1, 2, \dots, N - 1 \quad (3)$$

These synthetic indices are mapped into daily (24-hour) and seasonal (365-day) sine and cosine transformations.

- **Limitations:** When temporal gaps exist in the raw API archive, row-position-based encoding shifts the phase of the cyclical features relative to true local time. As a result, the model receives corrupted temporal signals (e.g., night hours encoded as daytime), breaking temporal integrity and reducing forecast accuracy.

4.3 Gap-aware preprocessing pipeline: LSTMv0.0.9-alpha

To address missing timesteps and irregular sampling without altering the underlying LSTM cell equations, LSTMv0.0.9-alpha introduces a gap-aware preprocessing pipeline featuring five domain-informed enhancements:

- **Fixed-grid reindexing:** The pipeline constructs an explicit datetime index with a strict 3-hour frequency across the entire archive range. The raw forecast archive is reindexed onto this grid, explicitly identifying missing timestamps and inserting blank rows.
- **Missingness mask:** A binary indicator feature vector $m_t \in \{0, 1\}^D$ is appended to the input, where $m_{t,d} = 0$ denotes an observed data point and $m_{t,d} = 1$ denotes an imputed entry. This allows the LSTM gates to distinguish real observations from synthetic values.
- **Delta-time feature (Δt):** The elapsed time since the last valid observation is computed and included as an auxiliary input feature. For normal 3-hour steps, $\Delta t = 3$; across missing gaps, Δt increases monotonically (e.g., 6h, 9h, 12h), providing explicit gap awareness to the network.
- **Physically-aware interpolation:** Missing values are imputed using domain-specific rules:
 - *General meteorological variables:* Standard linear interpolation is applied to continuous physical variables including temperature, humidity, wind speed, atmospheric pressure, and cloudiness.
 - *Solar radiance constraint:* Solar radiance is interpolated separately and strictly forced to zero during non-daylight hours (based on sun position). This prevents unphysical nighttime solar values that would otherwise result from standard linear interpolation across overnight gaps.
- **Large-gap window filtering:** Sequence windows containing an unobserved gap exceeding 12 consecutive hours are discarded entirely from training. This prevents the network from learning on heavily artificial segments where linear interpolation fails to capture rapid weather transitions.

5 Experiment Results

Evaluation metrics include test loss (Huber loss), margin of error calculation using Mean Absolute Percentage Error (MAPE), and improvement over naïve using Mean Absolute Error (MAE).

5.1 Dataset (1): complete data evaluation

When evaluated on the complete 346-day dataset without missing rows, both models yield virtually identical test performance.

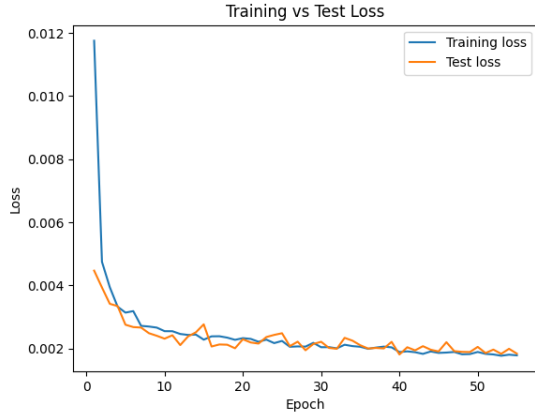
Table 1: Performance comparison on Dataset (1) — complete data (346 days, 0 missing rows).

Model variant	Test loss (Huber)	MAPE (%)	MAE
LSTMv0.0.9-beta (Baseline)	0.0018	2.90%	56.7%
LSTMv0.0.9-alpha (Gap-Aware)	0.0018	2.92%	56.3%

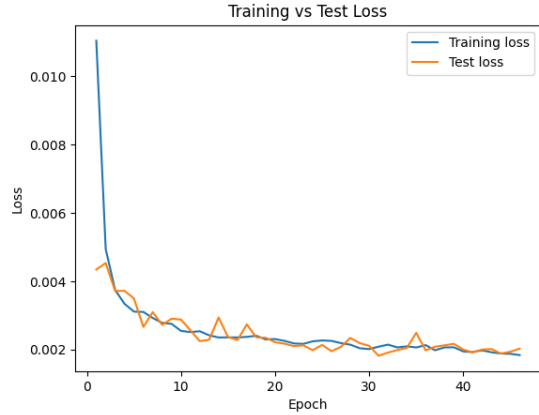
Analysis: On uninterrupted datasets, the gap-aware pipeline achieves an identical test loss of 0.0018 and a comparable MAPE (2.90% vs. 2.92%), with Alpha obtaining a marginal improvement in MAE (56.3% vs. 56.7%). These results confirm that on complete datasets, explicit gap-handling mechanisms provide little added value over row-position-based cyclical encoding.

5.2 Dataset (2): sparse data evaluation

Under real-world conditions with scattered missing timestamps, the gap-aware pipeline demonstrates a clear performance advantage over the naïve baseline.



(a) LSTMv0.0.9-beta on Dataset (1)



(b) LSTMv0.0.9-alpha on Dataset (1)

Figure 2: Training and test loss curves on Dataset (1) (complete data). Left: LSTMv0.0.9-beta. Right: LSTMv0.0.9-alpha.

Table 2: Performance comparison on Dataset (2) — sparse data (645 days, 86 missing rows).

Model variant	Test loss (Huber)	MAPE (%)	MAE
LSTMv0.0.9-beta (Baseline)	0.0014	2.59%	63.5%
LSTMv0.0.9-alpha (Gap-Aware)	0.0012	2.33%	67.0%

Analysis: As training data volume increases from 346 days to 645 days, both models benefit from the larger dataset size and achieve lower test losses than on Dataset (1). However, LSTMv0.0.9-alpha significantly outperforms Beta on sparse data, reducing test loss from 0.0014 to 0.0012 and MAPE from 2.59% to 2.33%. This demonstrates that Alpha successfully extracts more meaningful predictive signal from incomplete temporal sequences by preserving time step alignment through reindexing, delta-time, and missingness indicators.

6 Discussion and Limitations

6.1 Discussion of empirical findings

- **Temporal integrity vs. phase shift:** The primary failure mode of naive baseline models (Beta) on sparse data stems from row-position-based cyclical encoding. When timestamps are dropped without reindexing, row index sequence mappings shift local time forward, misaligning daily and seasonal sine/cosine features relative to true astronomical time. By enforcing a strict 3-hour grid, LSTMv0.0.9-alpha prevents phase distortion and maintains true temporal alignment.
- **Sensitivity to preprocessing quality:** While Alpha achieves superior accuracy on sparse archives, empirical logs indicate that it exhibits higher variance between repeated training runs compared to Beta. This higher variance is attributed to its dependence on preprocessing consistency, especially linear interpolation across missing intervals and non-daylight solar zero-forcing.
- **Practical deployment feasibility:** Unlike complex architectures such as GRU-D (which alters recurrent cell gating equations with trainable decays) or CRIB (which employs unified-attention and Information Bottleneck bounds without imputation), LSTMv0.0.9-alpha achieves gap awareness entirely through lightweight feature engineering. This makes Alpha directly compatible with standard deep learning production frameworks.

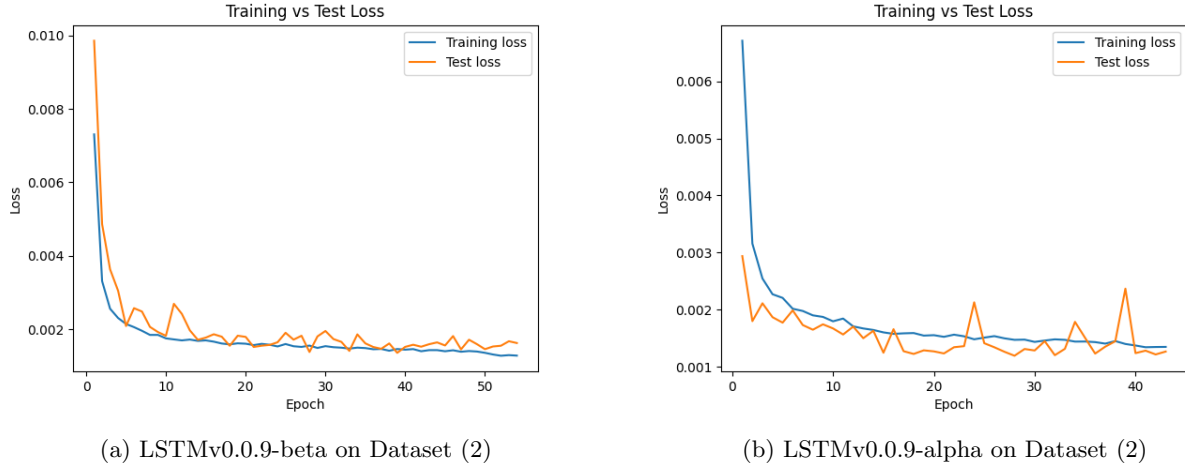


Figure 3: Training and test loss curves on Dataset (2) (sparse data). Left: LSTMv0.0.9-beta. Right: LSTMv0.0.9-alpha.

6.2 Limitations and future work

Despite its effectiveness, several limitations of this study should be noted:

- **Target and horizon scope:** The current pipeline is restricted to single-step point forecasts for temperature. It does not perform multi-target joint forecasting (e.g., temperature, humidity, and wind speed simultaneously) or multi-step extended horizon prediction (e.g., 3h to 24h+).
- **Interpolation artifacts:** Non-solar meteorological variables rely on standard linear interpolation across missing intervals. For gaps approaching the 12-hour filtering threshold, linear interpolation may smooth out sudden weather transitions or extreme local fluctuations.
- **Architecture comparison:** The study focuses exclusively on standard LSTM backbones. Future iterations should benchmark gap-aware pipelines against modern attention-based sequence models, Transformer architectures, and specialized trainable decay mechanisms (such as GRU-D and CRIB).
- **Geographic localization:** Experimental evaluation is currently limited to archived meteorological data from Ho Chi Minh City. Testing across diverse climate zones with varying weather volatility is required to confirm global generalization.

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The checklist follows the NeurIPS 2026 guidelines.

1. Claims

- Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?
- Answer: [Yes]
- Justification: The abstract and introduction explicitly describe the empirical nature of the study, the dataset collected from Ho Chi Minh City, the architectural choices of LSTMv0.0.9-beta vs. LSTMv0.0.9-alpha, and the performance metrics achieved.

2. Limitations

- Question: Does the paper discuss the limitations of the work performed by the authors?
- Answer: [Yes]
- Justification: Section 6 contains a dedicated discussion on study limitations, including the focus on single-step temperature forecasting, dependence on linear interpolation across large missing intervals, and higher run-to-run variance.

3. Theory Assumptions and Proofs

- Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
- Answer: [N/A]
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4. Experimental Result Reproducibility

- Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper?
- Answer: [Yes]
- Justification: Section 3 and Section 4 fully detail the 80/20 train/test split, 72-timestep lookback sequence length, input feature sets, Huber loss, Adam optimizer setup, and LSTM hyperparameters. Full source code is open-sourced.

5. Open Access to Data and Code

- Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?
- Answer: [Yes]
- Justification: The complete source code repository is published on GitHub under the Apache-2.0 license, and the curated dataset is released on Kaggle.

6. Experimental Setting/Details

- Question: Does the paper specify all the training and test details necessary to understand the results?
- Answer: [Yes]

- Justification: Data preprocessing steps (reindexing grid, missingness mask, solar zero-forcing, large-gap window filtering) and optimization parameters are fully specified in Section 3 and Section 4.

7. Experiment Statistical Significance

- Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?
- Answer: [No]
- Justification: Performance is reported via point metrics (test loss, MAPE, MAE) across two distinct 346-day and 645-day benchmark splits. Incorporating multi-seed error bars is noted as part of future work.

8. Experiments Compute Resources

- Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?
- Answer: [Yes]
- Justification: The models utilize a lightweight 2-layer LSTM architecture (`hidden_size = 128`), which can be trained on standard consumer-grade GPU or CPU hardware within short execution times.

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- Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?
- Answer: [Yes]
- Justification: The research uses publicly available API meteorological archives and open datasets, raising no ethical concerns regarding privacy or human subjects.

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- Answer: [Yes]
- Justification: Section 1 highlights positive societal impacts on urban planning and climate risk management in tropical cities. The potential for negative dual-use or harm is minimal due to the public nature of weather data.

11. Safeguards

- Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?
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- Justification: Weather forecasting models and public climate data do not pose dual-use safety or security risks requiring restricted access safeguards.

12. Licenses for Existing Assets

- Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned?
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- Justification: OpenWeatherMap API data sources are properly cited, and the codebase is released under the Apache-2.0 license.

13. New Assets

- Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?
- Answer: [Yes]
- Justification: The curated 647-day Ho Chi Minh City weather dataset and LSTMv0.0.9-alpha preprocessing scripts are fully documented in the repository README.md.

14. Crowdsourcing and Research with Human Subjects

- Question: For crowdsourcing experiments and research with human subjects, does the paper include full details?
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- Question: Does the paper describe potential risks incurred by study participants and IRB approvals?
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16. Declaration of LLM Usage

- Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods?
- Answer: [N/A]
- Justification: LLMs were not used as a component of the forecasting model or data processing pipeline.